

Little Explorers · Class 6

# Hot & Cold Temperature Light

Red means warm. Blue means cool!

Temperature sensor

Red and blue lights

Choosing a number

Did you know?

Your body stays at about 37 degrees, even on a cold winter day!



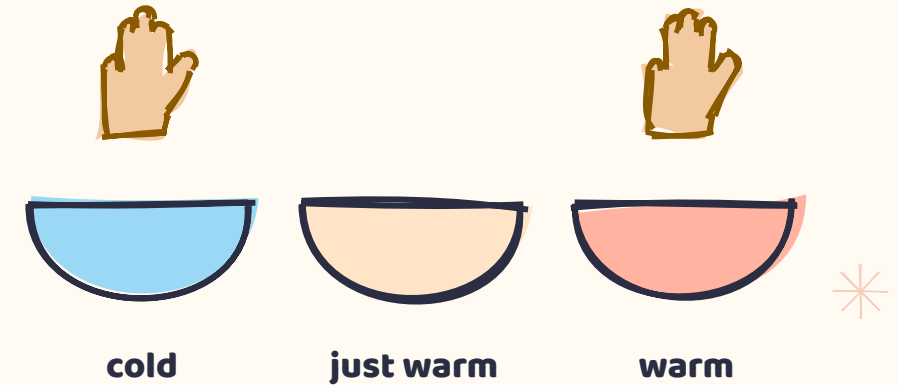
Let's wonder...

# Can our hands tell the truth?



## A tricky experiment

Put one hand in **cold** water and one hand in **warm** water. Wait.  
Now put **both** hands in the middle bowl.  
One hand says **warm**! The other says **cold**!



Our hands are confused!

Which hand is right?

Use a sensor!

**Today:** we build a machine that measures hot and cold with **numbers**, and shows it with a red or blue light.





## Learning aims

# Today I will learn to...



1



I can say if something is **hot** or **cold** using a **number**.

2



I can make a **red** or **blue** light switch on.

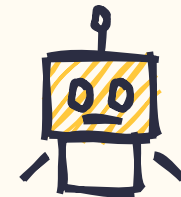
3



I can **choose the number** where red turns on.

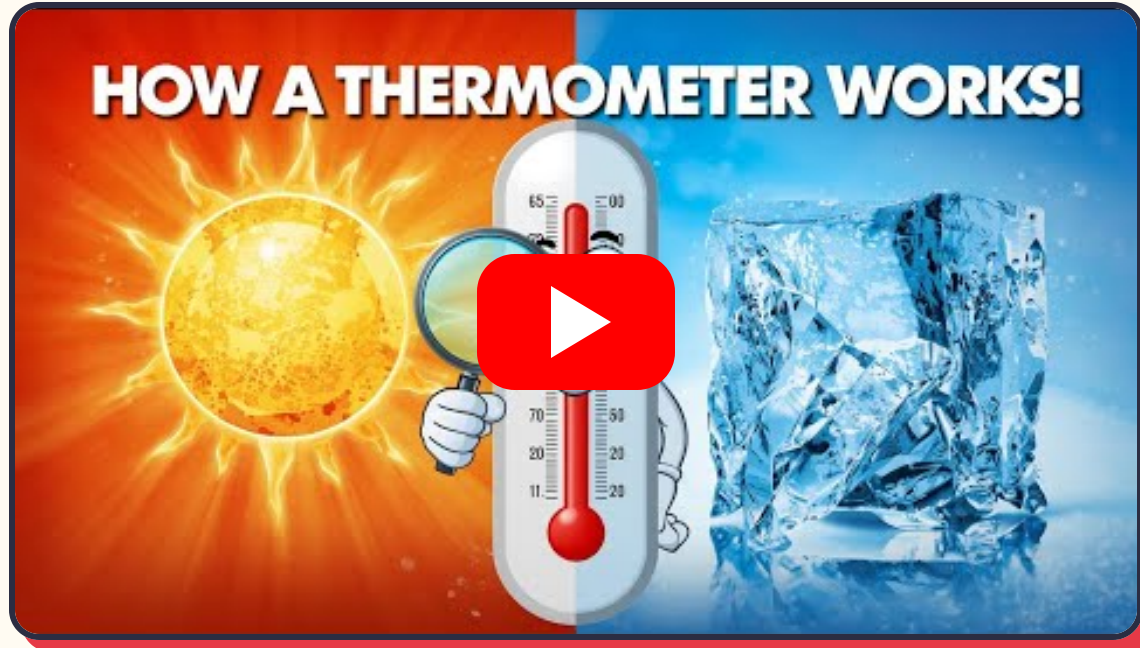
### Keywords

hot · cold · warm · cool · temperature  
· degrees · sensor



Watch and think

# Video time!



## How a Thermometer Works

Whimsy Wonders • about 5 minutes • [youtu.be/XYEX\\_2oxj9g](https://youtu.be/XYEX_2oxj9g)

### Watch for...

- 1 What does a **thermometer** measure?
- 2 What happens to the line when it gets **hot**?
- 3 Where do we use thermometers?



**After the video:** Show with your arm: arm up = hot, arm down = cold. Your partner says a place and you show it!



Hot and cold

# Why do we check temperature?



## Fever

Is my little brother poorly?  
Check with a thermometer



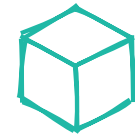
## Weather

Should I wear a jumper or a  
T-shirt today?



## Cooking

Is the milk too hot to drink?



## The fridge

Keeps food cold so it stays  
fresh



Where do YOU feel hot or cold? Kitchen? Car in summer? Under a fan?

## Ask at home

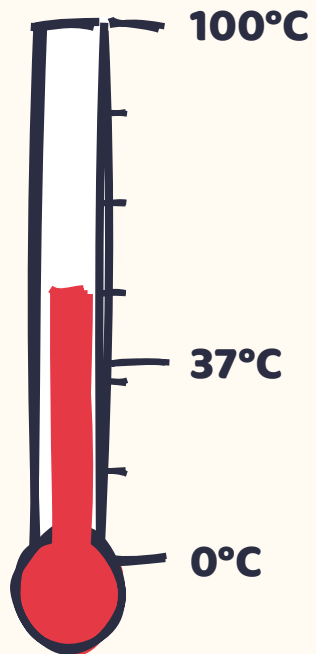
Do you have a thermometer? Where is it  
kept?





Big idea

# Temperature is a NUMBER



**100°C** boiling water (very hot!)



**37°C** inside our body



**0°C** ice

## Degrees

We write °C and say degrees. Bigger number = hotter!

## Try it

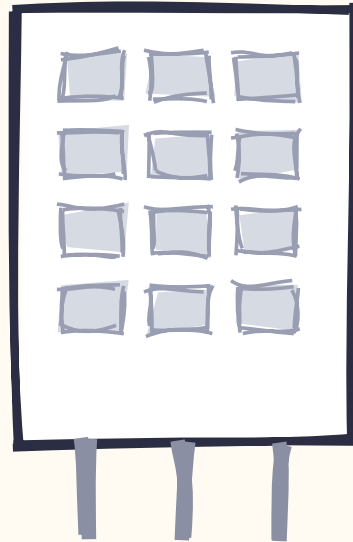
Is 30 hotter or colder than 20? What about 5?





Meet our helper

# The temperature sensor



## A tiny thermometer

This white box feels the air around it. It tells the Arduino the temperature as a **number**, like **27**.

## Fun fact

It can also tell if the air is damp (humidity). Like after rain!



## Keep it dry!

It is not waterproof. Never put it in water or in your mouth.



Class game

# Hot or cold?

## How to play

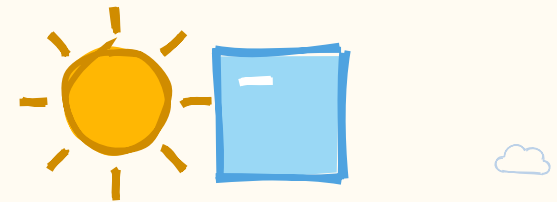
1. The teacher shows a picture card.
2. **HOT?** Stand up tall like the sun!
3. **COLD?** Curl up small like an ice cube!
4. Cards: ice cream, cup of tea, snowman, fire, the fridge, summer sun

## Then discuss

Is a cup of tea **always** hot? What happens if you wait one hour?

### Big idea

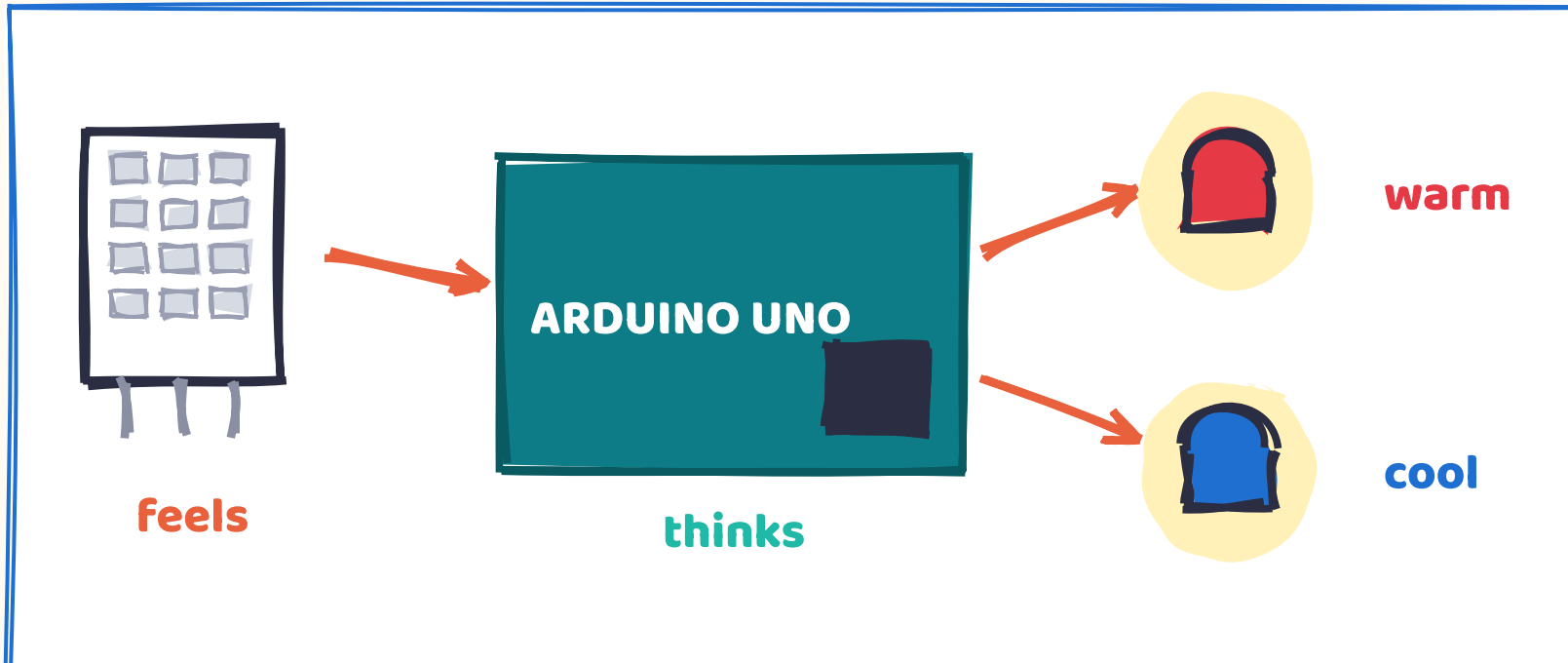
Hot things cool down and cold things warm up over time.





Our machine

# Feel, think, show



## How it works

The sensor **feels** the temperature.

The Arduino **thinks**: is it bigger than our number?

The lights **show** the answer.

**Teacher:** the sensor and two LEDs are pre-wired (data pin 2, red LED pin 12, blue LED pin 11).  
Children plug in, code and test.



Time to build!

# Let's code our light



when Arduino starts

forever

set temp to temperature (pin 2)

if temp &gt; 28

turn ON red light

turn OFF blue light

else

turn ON blue light

turn OFF red light

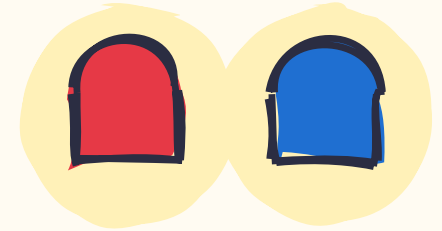
wait 2 seconds

## Read it like a story

Read the temperature.

IF it is more than 28: **red**.ELSE: **blue**.

Wait, then check again!



## Why wait 2 seconds?

Our sensor is a bit slow. It needs a little rest between readings.

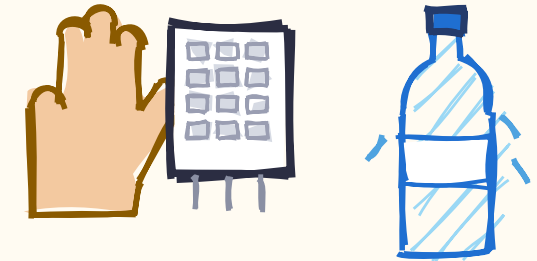




Experiment!

# Make it change colour

| Try this                     | I think...   | Number | Light |
|------------------------------|--------------|--------|-------|
| Leave it on the table        | red or blue? | _____  | _____ |
| Hold it gently in both hands | red or blue? | _____  | _____ |
| Hold it near a cold bottle   | red or blue? | _____  | _____ |
| Blow on it                   | red or blue? | _____  | _____ |



Be patient! The number changes **slowly**. Count to 20 before you look.

### Cold bottle tip

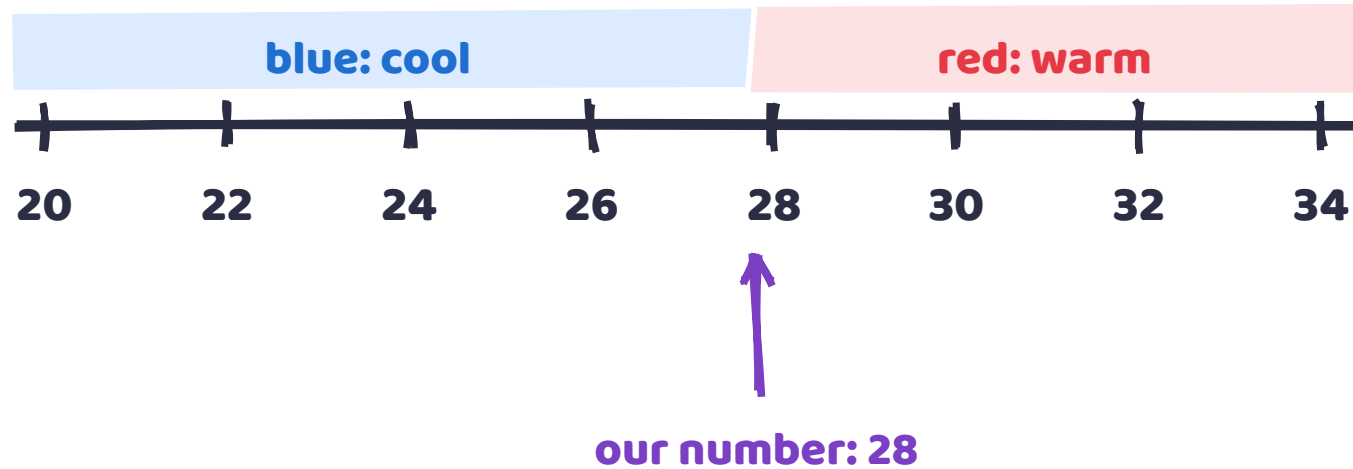
Wrap the bottle in a plastic bag so no water drips on the sensor.





You decide!

# Choose YOUR hot number



Bigger or smaller?

If the room is 27 and your number is 25, which light shows?

## Your turn

Change **28** in the code to a new number. Try **25**. Try **32**. When does the light change easily? When does it never go red?

Engineer word

The number where things change is called a **THRESHOLD**.





Explorers!

# Where is the warmest place?



| Place               | Number   | Light      |
|---------------------|----------|------------|
| Near the window     | _____ °C | red / blue |
| Under the fan       | _____ °C | red / blue |
| Near the door       | _____ °C | red / blue |
| Inside a school bag | _____ °C | red / blue |

## Talk about it

Which place was **warmest**?  
Which was **coolest**? Why do you think so?



## Teacher

Carry the laptop with the Arduino on a tray.  
Walk slowly together!





Stay safe

# Hot things can hurt!



**Never touch** a stove, kettle, iron or hot pan.



**Never put** our sensor in water, tea or your mouth.



**Ask a grown-up** to check if food or drinks are too hot.



**Cool a burn** with cool running water and tell a grown-up straight away.



## Remember

Our sensor measures the AIR. It is not for checking a fever.





Level up!

# Hot and cold challenges



## Choose your challenge



**Just right:** add a green light for a middle number, not too hot and not too cold.



**Beep when hot:** make the buzzer beep when the red light comes on.



**Blink fast:** make the red light blink when it is VERY hot.



**Class chart:** write everyone's hot number on the board. Whose was the biggest?

## Think like an engineer

Why do you think a fridge has a light that switches on when you open the door?



## Boss level

Make the light change at TWO numbers: blue, green, then red!





Real heroes

# Thermometers keep us safe



## Doctors

Check if you have a fever. 38°C or more means tell a grown-up!



## Fridges

Keep food cold so germs do not grow



## Farmers

Protect plants from frost and heat



If you could put a temperature light anywhere at home, **where** would you put it?

Draw it!

Draw your invention and show the class.





Show what you know

# What did we learn?

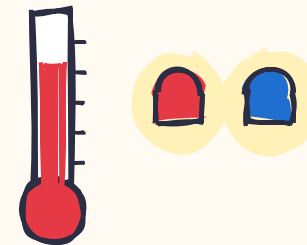
- ☒ Temperature is a number in degrees (°C)
- ☒ Bigger number = hotter
- ☒ Our sensor is a tiny thermometer for the Arduino
- ☒ IF it is hot: red. ELSE: blue
- ☒ We chose our own hot number (threshold)



**Exit ticket:** is 35°C hotter or colder than 20°C?

At home

With a grown-up, look at a weather app. What is today's temperature?



**Next class:** a  
Plant Water  
Alarm!

